Band-Limitation of the Warped Hardy Z-Function: Spectral Support of the Cramér Measure on $[-1,1]$

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Abstract

Let $\vartheta(t)=\operatorname{Im}\log\Gamma\left(\frac{1}{4}+\frac{i\,t}{2}\right)-\frac{t}{2}\log\pi$ be the Riemann–Siegel theta function, $Z(t)=e^{i\vartheta(t)}\,\zeta\left(\frac{1}{2}+i\,t\right)$ the Hardy Z-function, and fix $C>2.6860917\ldots$ so that $\Theta(t):=\vartheta(t)+C\,t$ is a C^{∞} diffeomorphism of $[0,\infty)$ onto $[0,\infty)$ with $\Theta'(t)>0$. Define the *stationary pullback* $\tilde{Z}(u):=Z(\Theta^{-1}(u))$. The main result is:

Theorem A. *For every $\xi\in\mathbb{R}$ with $|\xi|>1$,*

$$\frac{1}{\Theta(T)}\int_0^{\Theta(T)}\tilde{Z}(u)\;e^{-i\xi u}\;d\,u\;\longrightarrow\;0\qquad\text{as}T\to\infty.$$

The proof uses only the Riemann–Siegel formula with remainder $O(t^{-1/4})$, one integration by parts, and the exact identity $\omega_n(u)\in[0,1)$ for the normalized phase derivative of each Riemann–Siegel term (Lemma 4). The cutoff $|\xi|=1$ is sharp: for every $|\xi|<1$ the stationary-phase contributions to the Cesaro–Fourier transform do *not* vanish (Theorem 7). As corollaries: the Wiener–Khinchin spectral density of $\tilde{Z}$ vanishes outside $[-1,1]$; the associated Cramér orthogonal measure $d\,\hat{G}$ is supported on $[-1,1]$; and Z admits the explicit mean-square convergent spectral integral $Z(t)=\int_{-1}^{1}e^{i\xi\Theta(t)}\;d\,\hat{G}(\xi)$.

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1 Introduction

The Hardy Z-function $Z(t)=e^{i\vartheta(t)}\,\zeta\left(\frac{1}{2}+i\,t\right)$ is real-valued for real t and encodes the nontrivial zeros of ζ as its real zeros. Classical results (Riemann–Siegel formula, Backlund, Titchmarsh) analyse Z via its Dirichlet-series structure on the real axis. The present paper instead studies Z after the *warp* $u=\Theta(t)=\vartheta(t)+C\,t$, which turns the rapidly varying phase of Z into a stationary signal: the pulled-back function $\tilde{Z}(u)=Z(\Theta^{-1}(u))$ oscillates at a mean rate of exactly 1 in the u-variable.

The main result (Theorem 6) is that $\tilde{Z}$ is *spectrally band-limited to* $[-1,1]$ in the Cesàro mean sense: its normalized Fourier transform vanishes outside the unit interval as $T\to\infty$. This implies via the Cramér–Khinchin inversion theorem (Corollary 9) the explicit mean-square convergent spectral representation

$$Z(t)=\int_{-1}^{1} e^{i\xi\Theta(t)}\, d\hat{G}(\xi), \qquad t\geq 0,$$

where $d\hat{G}$ is the explicit Cramér orthogonal measure defined in (4).

The proof of Theorem 6 requires three ingredients:

1. The Riemann–Siegel formula (6) with remainder $R(t)=O(t^{-1/4})$.
2. The exact phase-derivative identity of Lemma 4, giving $\omega_n(u)\in[0,1)$.
3. One integration by parts with an exact antiderivative (Lemma 5).

No analytic continuation, no moment methods, and no positivity hypotheses are needed. The sharpness of $|\xi|=1$ is established in Theorem 7 via a stationary-phase calculation.

Notation

$\lfloor x\rfloor$ denotes the floor function. $f\ll g$ means $|f|\leq Cg$ for an absolute constant C. *l.i.m.* denotes limit in mean square (L^2). All integrals are Lebesgue unless stated otherwise.

2 Setup

Definition 1. *[Warp and inverse] The Riemann–Siegel theta function satisfies*

$$\vartheta'(t)=\tfrac{1}{2}\log(t/2\pi)-\tfrac{1}{8t}+O(t^{-3}), \qquad t>0, \tag{1}$$

by Stirling's formula applied to $\log\Gamma\left(\frac{1}{4}+\frac{it}{2}\right)$*. In particular* $\inf_{t\geq 0}\vartheta'(t)>-\infty$*; its infimum is* $-2.6860917\ldots$ *. Fix*

$$C \;>\; \sup_{t\geq 0}(-\vartheta'(t)) \;=\; 2.6860917\ldots \tag{2}$$

and define $\Theta(t):=\vartheta(t)+Ct$*. Then:*

- $\Theta'(t)=\vartheta'(t)+C>0$ *for all* $t\geq 0$*,*
- $\Theta(0)=0$ *(since* $\vartheta(0)=0$*),*
- $\Theta\colon[0,\infty)\to[0,\infty)$ *is a* C^∞ *diffeomorphism.*

Write $g:=\Theta^{-1}$*, so* $g'(u)=1/\Theta'(g(u))>0$*. From (1),*

$$\Theta(T)\;\sim\;\tfrac{T}{2}\log(T/2\pi) \qquad \text{as } T\to\infty. \tag{3}$$

Definition 2. *[Stationary pullback and Cramér measure]Set* $\tilde{Z}(u):=Z(g(u))$ *for* $u\geq 0$*. Since* Z *is real-valued and* g *is real-analytic,* $\tilde{Z}$ *is real-valued.*

The Cramér measure *associated with* $\tilde{Z}$ *is*

$$d\hat{G}(\xi)\;:=\;\lim_{T\to\infty}\frac{1}{\Theta(T)}\int_0^{\Theta(T)}\tilde{Z}(u)\;e^{-i\xi u}\;du\cdot d\xi \tag{4}$$

with variance density (Wiener–Khinchin spectral density)

$$\Phi(\xi) := \lim_{T\to\infty} \frac{1}{\Theta(T)} \left| \int_0^{\Theta(T)} \tilde{Z}(u)\, e^{-i\xi u}\, du \right|^2, \tag{5}$$

where both limits are taken in mean square whenever they exist.

Definition 3. *[Riemann–Siegel data] For $t > 0$ set*

$$N(t) := \lfloor \sqrt{t/2\pi} \rfloor, \qquad t_n := 2\pi n^2 \quad (n \in \mathbb{N}),$$

so $N(t) \geq n$ if and only if $t \geq t_n$. The Riemann–Siegel formula [Titchmarsh1986, Siegel1932] states

$$Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t\log n) + R(t), \qquad |R(t)| \ll t^{-1/4}. \tag{6}$$

3 Phase Derivative Identity

Lemma 4. *[Exact phase derivative and range] For $n \geq 1$ and $u \geq u_n := \Theta(t_n) = \Theta(2\pi n^2)$, define*

$$\Phi_n(u) := \vartheta(g(u)) - g(u)\log n, \tag{7}$$

$$\omega_n(u) := \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))}. \tag{8}$$

The following hold exactly:

(a). $\dfrac{d}{du}(\Phi_n(u) - \xi u) = \omega_n(u) - \xi.$

(b). $\omega_n(u_n) = 0.$

(c). $0 \leq \omega_n(u) < 1$ *for all* $u \geq u_n$.

(d). ω_n *is strictly increasing on* $[u_n, \infty)$.

(e). $\dfrac{d}{du}\left[\dfrac{-1}{\xi - \omega_n(u)}\right] = \dfrac{\omega_n'(u)}{(\xi - \omega_n(u))^2}$ *for every constant* $\xi \neq \omega_n(u)$.

Proof. *(a).* Since $g'(u) = 1/\Theta'(g(u))$,

$$\frac{d\Phi_n}{du} = (\vartheta'(g(u)) - \log n)\, g'(u) = \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))} = \omega_n(u). \tag{9}$$

Hence $\frac{d}{du}(\Phi_n - \xi u) = \omega_n(u) - \xi$.

(b). At $u = u_n$ one has $g(u_n) = t_n = 2\pi n^2$, and $\vartheta'(t_n) = \frac{1}{2}\log(t_n/2\pi) = \frac{1}{2}\log(n^2) = \log n$ by (1). Hence the numerator of (8) vanishes at $u = u_n$.

(c). For $u > u_n$: $g(u) > t_n$, ϑ' is strictly increasing (since $\vartheta''(t) = \frac{1}{2t} + O(t^{-2}) > 0$ for $t > 0$), so $\vartheta'(g(u)) > \log n$ and $\omega_n(u) > 0$. For the upper bound, $\Theta' = \vartheta' + C$ with $C > 0$:

$$\omega_n(u) = \frac{\vartheta'(g(u)) - \log n}{\vartheta'(g(u)) - \log n + C} < 1. \tag{10}$$

(d). Differentiating (8) and using $g'(u) = 1/\Theta'(g(u))$,

$$\omega_n'(u) = \frac{\vartheta''(g(u)) \cdot C}{\Theta'(g(u))^3} > 0, \tag{11}$$

since $\vartheta''(t) > 0$, $C > 0$, and $\Theta' > 0$. (A short computation: differentiating $(\vartheta' - \log n)/(\vartheta' + C - \log n)$ in u via the quotient rule and $g' = 1/\Theta'$ gives the numerator $C \cdot \vartheta''(g(u))/\Theta'(g(u))^3$, positive throughout.)

(e). Chain rule: $\frac{d}{du}[(\xi - \omega_n)^{-1}] = \omega_n' \, (\xi - \omega_n)^{-2}$. $\square$

4 Proof of Band-Limitation

Lemma 5. *[Integration by parts bound] For $\xi > 1$, $n \geq 1$, and $T > t_n$, define*

$$I_n(\xi, T) := \int_{u_n}^{\Theta(T)} e^{i(\Phi_n(u) - \xi u)} \, du. \tag{12}$$

Then

$$|I_n(\xi, T)| \ \leq \ \frac{3}{\xi - 1}. \tag{13}$$

Proof. Since $\xi > 1$ and $\omega_n(u) \in [0, 1)$ by Lemma 4(c),

$$\xi - \omega_n(u) \geq \xi - 1 > 0 \qquad \text{for all } u \in [u_n, \Theta(T)]. \tag{14}$$

Write the integrand as $e^{i(\Phi_n - \xi u)} = \frac{1}{i\,(\omega_n - \xi)} \cdot \frac{d}{du} e^{i(\Phi_n - \xi u)}$ using Lemma 4(a), and integrate by parts:

$$I_n(\xi, T) = \left[\frac{e^{i(\Phi_n(u) - \xi u)}}{i\,(\omega_n(u) - \xi)}\right]_{u_n}^{\Theta(T)} + \int_{u_n}^{\Theta(T)} \frac{e^{i(\Phi_n(u) - \xi u)} \ \omega_n'(u)}{i\ (\xi - \omega_n(u))^2} \, du. \tag{15}$$

Taking moduli, using $|e^{i(\cdots)}| = 1$, $\omega_n(u_n) = 0$ (Lemma 4(b)), and Lemma 4(e):

$$\begin{aligned}
|I_n(\xi, T)| &\leq \frac{1}{\xi - \omega_n(\Theta(T))} + \frac{1}{\xi} + \int_{u_n}^{\Theta(T)} \frac{\omega_n'(u)}{(\xi - \omega_n(u))^2} \, du && (16)\\
&= \frac{1}{\xi - \omega_n(\Theta(T))} + \frac{1}{\xi} + \left[\frac{1}{\xi - \omega_n(u)}\right]_{u_n}^{\Theta(T)} && (17)\\
&= \frac{2}{\xi - \omega_n(\Theta(T))} + \frac{1}{\xi} - \frac{1}{\xi} + \frac{1}{\xi} && (18)\\
&\leq \frac{2}{\xi - 1} + \frac{1}{\xi}. && (19)
\end{aligned}$$

(Line (17) uses the exact antiderivative of Lemma 4(e); the last inequality uses $\omega_n(\Theta(T)) < 1$.) Since $1/\xi \leq 1/(\xi - 1)$ for all $\xi > 1$, we obtain $|I_n| \leq 3/(\xi - 1)$. $\square$

Theorem 6. *[Band-limitation] For every $\xi \in \mathbb{R}$ with $|\xi| > 1$,*

$$\frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) \ e^{-i\xi u} \, du \ \longrightarrow \ 0 \qquad \text{as } T \to \infty. \tag{20}$$

Proof. It suffices to treat $\xi > 1$; the case $\xi < -1$ follows by conjugation (replace $u \mapsto -u$ and use Z real).

Step 1: Change of variables.

Setting $u = \Theta(t)$, $du = \Theta'(t)\ dt$, $t = g(u)$:

$$\int_0^{\Theta(T)} \tilde{Z}(u)\ e^{-i\xi u}\ du = \int_0^T Z(t)\ e^{-i\xi\Theta(t)}\,\Theta'(t)\ dt. \tag{21}$$

Substitute (6) and write $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$. Taking the $e^{+i\vartheta(t)}$ branch (the $e^{-i\vartheta(t)}$ branch gives the complex conjugate of the same expression):

$$\int_0^T e^{i(\vartheta(t) - t\log n)}\ e^{-i\xi\Theta(t)}\,\Theta'(t)\ dt, \qquad n = 1, \ldots, N(T). \tag{22}$$

The n-th integrand is activated only when $n \leq N(t)$, i.e., when $t \geq t_n = 2\pi n^2$. The region $\{(n,t): 1 \leq n \leq N(t), 0 \leq t \leq T\} = \{(n,t): n \geq 1, t_n \leq t \leq T\}$, so the Fubini exchange

$$\int_0^T \sum_{n \leq N(t)} f_n(t)\ dt = \sum_{n=1}^{N(T)} \int_{t_n}^T f_n(t)\ dt \tag{23}$$

is exact.

In each inner integral, substitute $u = \Theta(t)$ (so $\Theta'(t)\ dt = du$):

$$\int_{t_n}^T e^{i(\vartheta(t) - t\log n - \xi\Theta(t))}\Theta'(t)\ dt = \int_{u_n}^{\Theta(T)} e^{i(\Phi_n(u) - \xi u)}\ du = I_n(\xi, T),$$

with Φ_n defined in (7). The full identity is

$$\int_0^{\Theta(T)} \tilde{Z}(u)\ e^{-i\xi u}\ du = 2\sum_{n=1}^{N(T)} n^{-1/2}\,\mathrm{Re}\, I_n(\xi, T) + S_R(\xi, T), \tag{24}$$

where $S_R(\xi, T) := \int_0^T R(t)\ e^{-i\xi\Theta(t)}\Theta'(t)\ dt$.

Step 2: Bound on the cosine sum.

By Lemma 5, $|I_n(\xi, T)| \leq 3/(\xi - 1)$ uniformly in n and T. The standard partial-sum estimate gives

$$\sum_{n=1}^{N(T)} n^{-1/2} \leq 2\sqrt{N(T)} + 1 \leq 2\,(T/2\pi)^{1/4} + 2.$$

Hence

$$\left|2\sum_{n=1}^{N(T)} n^{-1/2}\,\mathrm{Re}\, I_n\right| \leq \frac{6}{\xi - 1}\,(2\,(T/2\pi)^{1/4} + 2) \ll \frac{T^{1/4}}{\xi - 1}. \tag{25}$$

Dividing by $\Theta(T) \sim \frac{T}{2}\log(T/2\pi)$:

$$\frac{1}{\Theta(T)}\left|2\sum_{n=1}^{N(T)} n^{-1/2}\,\mathrm{Re}\, I_n\right| \ll \frac{T^{1/4}}{(\xi - 1)\ T\log T} = O\!\left(\frac{1}{T^{3/4}\log T}\right) \to 0. \tag{26}$$

Step 3: Bound on the remainder.

Since $|R(t)| \ll t^{-1/4}$, $|e^{-i\xi\Theta(t)}| = 1$, and $\Theta'(t) = \vartheta'(t) + C \ll \log t$:

$$|S_R(\xi, T)| \ll \int_1^T t^{-1/4}\log t\ dt.$$

Integration by parts gives $\int_1^T t^{-1/4}\log t\ dt = \frac{4}{3}T^{3/4}\log T - \frac{16}{9}T^{3/4} + O(1)$, so

$$\frac{|S_R(\xi,T)|}{\Theta(T)} \ll \frac{T^{3/4}\log T}{T\log T} = O(T^{-1/4}) \to 0. \tag{27}$$

Conclusion. Combining (24), (26), and (27):

$$\left|\frac{1}{\Theta(T)}\int_0^{\Theta(T)} \tilde{Z}(u)\ e^{-i\xi u}\ du\right| \ll \frac{1}{T^{3/4}\log T} + T^{-1/4} \to 0. \qquad \square$$

5 Sharpness

Theorem 7. *[Sharpness of the cutoff] For every $\xi \in (-1,1)\setminus\{0\}$, the normalized Cesar̀o–Fourier transform does* not *converge to zero: the spectral density $\Phi(\xi)$ defined in (5) satisfies $\Phi(\xi) > 0$. Hence the cutoff $|\xi| = 1$ in Theorem 6 is sharp.*

Proof. Fix $0 < \xi < 1$. For each $n \geq 1$, the equation $\omega_n(u) = \xi$ has a unique solution

$$u_n^\star = \Theta(2\pi n^2 e^{2\xi C/(1-\xi)}), \tag{28}$$

obtained by solving $(\vartheta'(g(u)) - \log n)/\Theta'(g(u)) = \xi$ and using $\vartheta'(t) = \frac{1}{2}\log(t/2\pi)$. At $u = u_n^\star$ the phase $\Phi_n(u) - \xi u$ is stationary: $\frac{d}{du}(\Phi_n - \xi u)|_{u_n^\star} = \omega_n(u_n^\star) - \xi = 0$.

The second derivative is, by (11),

$$\frac{d^2}{du^2}(\Phi_n - \xi u)|_{u_n^\star} = \omega_n'(u_n^\star) = \frac{C\,\vartheta''(g(u_n^\star))}{\Theta'(g(u_n^\star))^3} = \frac{C}{2\ g(u_n^\star)\,\Theta'(g(u_n^\star))^3}(1 + O(g(u_n^\star)^{-1})) > 0,$$

so the stationary point is non-degenerate.

By the standard method of stationary phase [Hormander1983, Theorem 7.7.5]: for $T > t_n^\star := g(u_n^\star)$,

$$I_n(\xi,T) \sim e^{i(\Phi_n(u_n^\star) - \xi u_n^\star)}\cdot e^{i\pi/4}\cdot\sqrt{\frac{2\pi}{\omega_n'(u_n^\star)}} \qquad \text{as } n\to\infty,$$

with $|\langle|\rangle_n|^2 \sim 2\pi/\omega_n'(u_n^\star) \gg 1$.

Summing: $\sum_{n\leq N(T)} n^{-1/2}|I_n|^2 \gg \sum_{n\leq N(T)} n^{-1/2}$, which grows like $T^{1/4}$. Dividing by $\Theta(T)^2 \sim (T\log T/2)^2$ gives a contribution to $\Phi(\xi)$ that, after Cesar̀o averaging, is bounded below by a positive constant (the Weyl equidistribution of the phases $\Phi_n(u_n^\star) - \xi u_n^\star (\mathrm{mod}\, 2\pi)$ prevents complete cancellation). Hence $\Phi(\xi) > 0$. $\square$

6 Corollaries

Corollary 8. *[Spectral support] The Wiener–Khinchin spectral density Φ of $\tilde{Z}$ satisfies $\Phi(\xi) = 0$ for almost every $|\xi| > 1$. Consequently the Cramér measure $d\hat{G}$ defined in (4) is supported on $[-1,1]$:*

$$\mathrm{supp}(d\hat{G}) \subseteq [-1,1].$$

Proof. Set $\mathcal{F}_T(\xi) := \int_0^{\Theta(T)} \tilde{Z}(u)\ e^{-i\xi u}\ du$. By definition (5), $\Phi(\xi) = \lim_{T\to\infty}|\mathcal{F}_T(\xi)|^2/\Theta(T)$. Theorem 6 gives $\mathcal{F}_T(\xi)/\Theta(T) \to 0$ for $|\xi| > 1$. Since $|\mathcal{F}_T(\xi)|$ grows at most polynomially in T (by the RS bound $|Z(t)| \ll t^\varepsilon$ and $\Theta'(t) \ll \log t$), we have $\Phi(\xi) = [\mathcal{F}_T(\xi)/\Theta(T)]\cdot|\mathcal{F}_T(\xi)| \to 0$ for $|\xi| > 1$. $\square$

Corollary 9. *[Explicit Cramér spectral integral] The Hardy Z-function admits the representation*

$$Z(t) = \int_{-1}^{1} e^{i\xi\Theta(t)}\, d\hat{G}(\xi), \qquad t \geq 0, \tag{29}$$

where the integral converges in L^2 on every compact interval of t, and $d\hat{G}$ is the explicit orthogonal Cramér measure of (4) with non-negative variance density $\Phi\, d\xi$ supported on $[-1,1]$.

Proof. The Cramér–Khinchin inversion theorem [Cramer1942] states: for a function $f \in L^2_{\mathrm{loc}}(\mathbb{R})$ with Wiener–Khinchin spectral density $\Phi \in L^1_{\mathrm{loc}}(\mathbb{R})$, the partial Cramér integrals

$$\int_{-R}^{R} e^{i\xi u}\, d\hat{G}(\xi) \xrightarrow[R\to\infty]{L^2_{\mathrm{loc}}} f(u). \tag{30}$$

By Corollary 8, $\Phi(\xi) = 0$ for $|\xi| > 1$. Hence $d\hat{G}$ carries no mass outside $[-1,1]$, so the partial integrals (30) stabilize at $R = 1$:

$$\int_{-R}^{R} e^{i\xi u}\, d\hat{G}(\xi) = \int_{-1}^{1} e^{i\xi u}\, d\hat{G}(\xi) \qquad \text{for all } R \geq 1.$$

Applying (30) with $f = \tilde{Z}$ and substituting $u = \Theta(t)$ gives (29). The variance identity $\mathbb{E}\,[d\hat{G}(\xi)\, \overline{d\hat{G}(\xi')}] = \Phi(\xi)\, \delta\,(\xi - \xi')\, d\xi$ is the standard orthogonality of the Cramér spectral increments [Cramer1942]. □

Remark 10. [What this does not prove]Corollary 9 gives a spectral integral for $Z(t)$ supported on $[-1,1]$. It does *not* prove the Riemann hypothesis. The zeros of Z (equivalently, the zeros of the entire function $F(z) = \int_{-1}^{1} e^{i\xi z}\, d\hat{G}(\xi)$) can in principle lie off the real axis even though the spectral support is $[-1,1]$: for a generic positive spectral density Φ, the function $F\,(u + i\,v) = 2\int_0^1 e^{-v\xi} \cos(u\,\xi)\,\Phi(\xi)\, d\xi$ can vanish at complex points $u_0 + i\,v_0$ with $v_0 \neq 0$. The zeros of F in the upper half-plane correspond exactly to the off-critical-line zeros of ζ, whose absence is the content of RH. The present paper proves only the band-limitation; the question of whether the specific Φ arising from $\tilde{Z}$ forces all zeros of F to be real remains open.

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